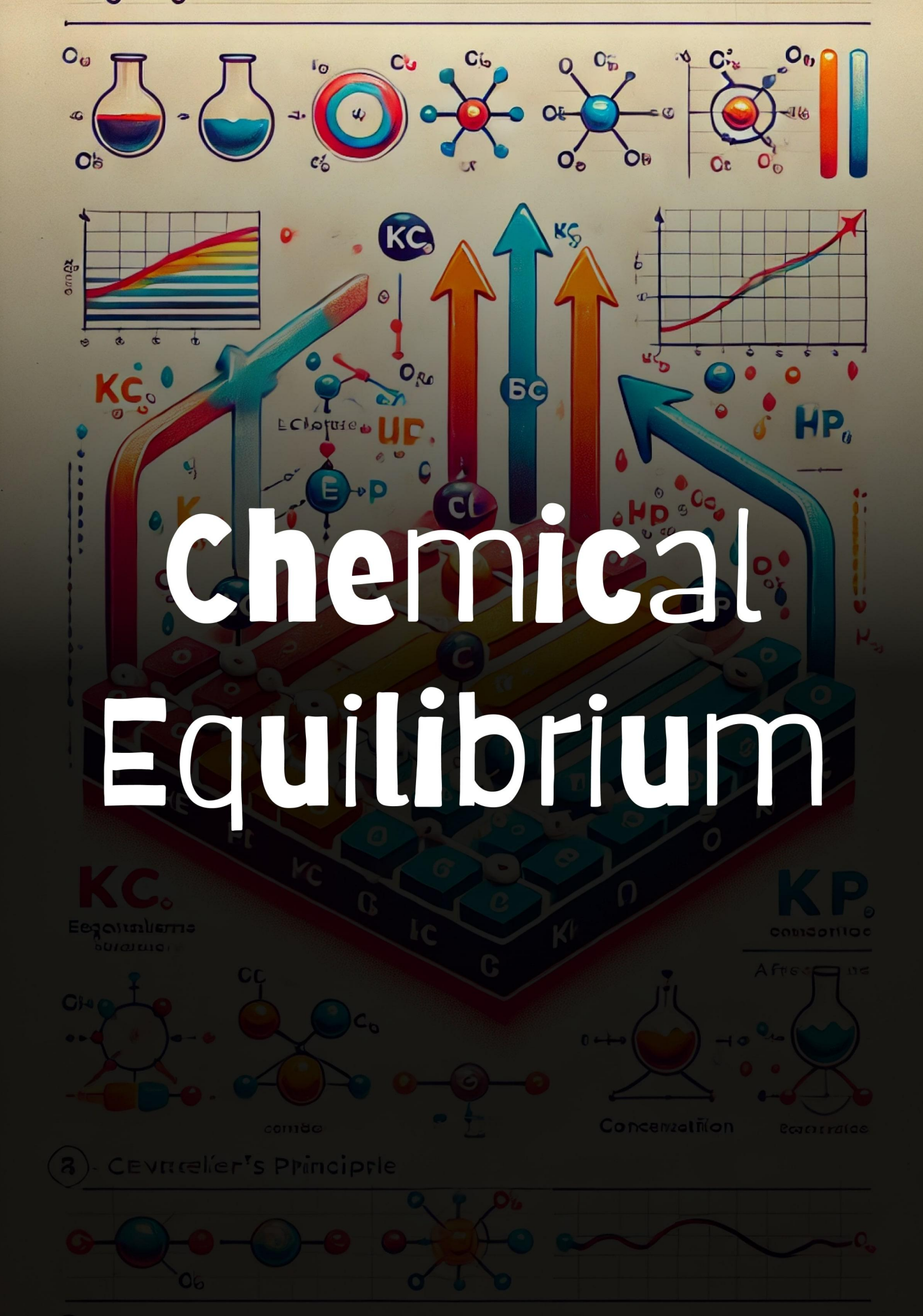

Chemical
Equilibrium

CHEMICAL EQUILIBRIUM

Reversible and unreversible reaction – Reaction can be classified into two types.

1. IRREVERSIBLE REACTION

The reactions in which the reactants are converted into products and products are not converted into the reactants again are called as **irreversible reactions.** These reactions are denoted by '$\longrightarrow$'.

An irreversible reaction always tends towards completion.

The following types of reactions are generally irreversible :–

1. THERMAL DECOMPOSITION

Decomposition of a substance in an open vessel is irreversible.

e.g :–

i. $CaCO_3 \xrightarrow{\Delta} CaO + CO_2\uparrow$

ii. $2KClO_3 \xrightarrow{\Delta} 2KCl + 3O_2\uparrow$

2. NEUTRALIZATION REACTION

Neutralization reaction between a strong acid and strong base is irreversible.

eg:- (i) $HCL + NaOH \longrightarrow NaCl + H_2O$

(ii) $H_2SO_4 + Ca(OH)_2 \longrightarrow CaSO_4 + H_2O$

3. PRECIPITATION REACTION

The Reaction between two electrolytes to form a precipitate is generally irreversible.

eg:- (i) $AgNO_3 + KCL \longrightarrow AgCl\downarrow + KNO_3$

4. COMBUSTION REACTION

Combustion of an organic substance is generally irreversible.

(2.) REVERSIBLE REACTION

The reaction in which reaction form products and products are again converted into reactants are called as reversible reaction

These reaction are denoted by a 'double headed arrow.'

$A \rightleftharpoons B$

The following types of reaction are general reversible :-

1. THERMAL DECOMPOSITION IN A CLOSED VESSEL

eg:- $CaCO_3 \rightleftharpoons CaO + CO_2$

$2KClO_3 \rightleftharpoons 2KCl + 3O_2$

2. NEUTRALIZATION REACTION

The reaction in which either acid or base or both are weak are reversible.

eg:- i $CH_3COOH + NaOH \rightleftharpoons CH_3COONa + H_2O$

(WA) (SB)

ii $HCl + NH_4OH \longrightarrow NH_4Cl + H_2O$

(SA) (WB)

iii $CH_3COOH + NH_4OH \rightleftharpoons CH_3COONH_4 + H_2O$

3. HYDROLYSIS REACTION

The hydrolysis reaction of salts of :-

Weak Base + Strong Acid

WA + SB

WA + WB

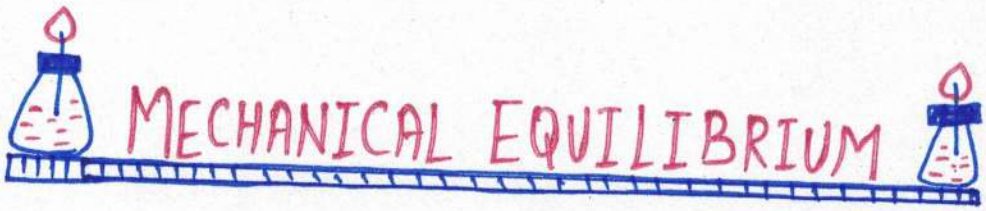
are reversible.

➤ Reversible reactions never tends to completion.

CONCEPT OF EQUILIBRIUM STATE

The state attained by a system in which the system has no tendancy for a change untill an external stimulation is given on the basis of behaviour of equilibrium state, it is classified into three types :-

1. MECHANICAL EQUILIBRIUM



a. Static Mechanical equilibrium

The Position or speed of a system with time, then it is called as mechanical equilibrium.

If the position of the system doesn't change with time, it is called static mechanical equilibrium.

CHARACTERISTICS OF CHEMICAL EQ^n

i. It can be obtained from either side of the reaction.

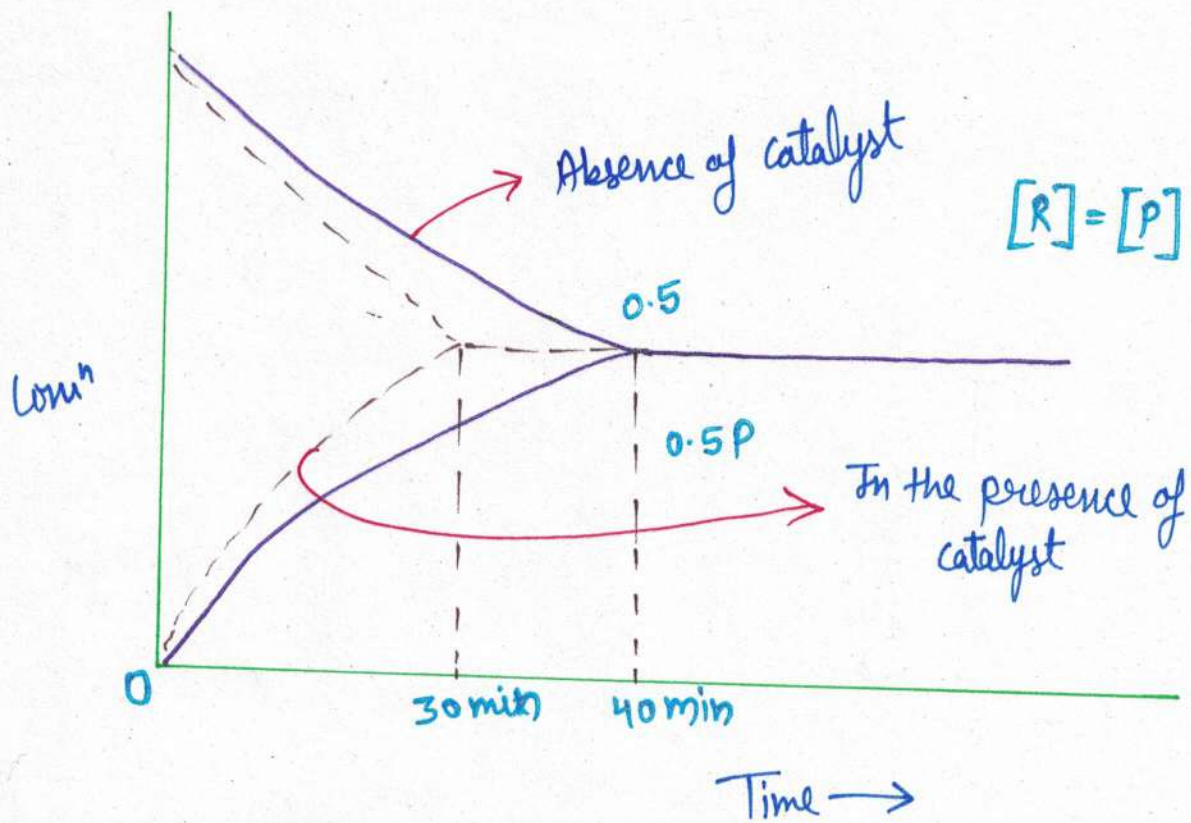
eg:- @ $2SO_3 \rightleftharpoons 2SO_2 + O_2$

$2SO_2 + O_2 \rightleftharpoons SO_3$

ii. Chemical equation is "dynamic" in nature, i.e:- the reaction takes place in both the directions with the same rate but it never stops.

iii. At eq^n, all measurable properties of system such as surface tension, viscocity, refraction index (μ), concentration, temperature pressure etc. do not change with time.

IV. A catalyst doesn't effect the eq^n state but in presence of a catalyst eq^m is achieved quickly as it increases weight of both forward and backward reaction to the same extent.



LAW OF MASS ACTION

- The rate of an elementary reaction (single-step) is directly proportional to product of active masses of reactants raised to a power equal to stichiometric co-efficient in a balanced chemical equation.

$$m_1A + m_2B \rightleftharpoons n_1C \quad n_2D$$

$$r_f \propto a_A^{m_1} \cdot a_B^{m_2}$$

- In a solution phase reaction

$$a_A \propto \text{Molarity} \Rightarrow a_A = \gamma[A]$$

$$a_A \propto \text{Mole fraction} \Rightarrow a_A = \gamma X_A$$

{Where γ and γ are constants & eq. to 1 for dil. Sol^n}

- For detail soluⁿ

$$a_A = [M] \quad \text{or} \quad a_A = X_A$$

- For gas phase reaction

$$a_A \propto P_A \Rightarrow a_A = f \cdot P_A$$

{Where $F=1$}

$a_{solids} = 1$

$a_{pure\ liquid} = 1$

$$a_A = P_A$$

$m_1A + m_2B \rightleftharpoons n_1C + n_2D$

Apply Law of Mass Action

$$r_F \propto [A]^{m_1} \cdot [B]^{m_2}$$

$$r_F = k_F [A]^{m_1} \cdot [B]^{m_2}$$

↓

Rate Const. of forward reaction

$$r_b \propto [C]^{n_1} \cdot [D]^{n_2}$$

$$r_b = k_b [C]^{n_1} \cdot [D]^{n_2}$$

↓

Rate Const. of backward reaction

$[\ ] \rightarrow$ Constant

At Equilibrium

$$r_F = r_b$$

$$k_F \cdot [A]_{eq}^{m_1} \cdot [B]_{eq}^{m_2} = k_b [C]_{eq}^{n_1} \cdot [D]_{eq}^{n_2}$$

$$K_c = \frac{k_F}{k_b} = \frac{[C]_{eq}^{n_1} \cdot [D]_{eq}^{n_2}}{[A]_{eq}^{m_1} \cdot [B]_{eq}^{m_2}}$$

↓

equilibrium constant

Write K_c constants for the foll reaction :-

i. $N_{2(g)} + 3H_{2(g)} \rightleftharpoons 2NH_{3(g)}$

ii. $2SO_{2(g)} + O_{2(g)} \rightleftharpoons 2SO_{3(g)}$

iii. $CaCO_{3(s)} \rightleftharpoons CaO_{(s)} + CO_{2(g)} \uparrow$

Ans i $K_c = \dfrac{[NH_3]^2}{[H_2]^3 . [N_2]^1} = \dfrac{[mol/L]^2}{[mol/L]^4} = [mol/L]^{-2}$

ii $K_c = \dfrac{[SO_3]^2}{[SO_2]^2 . [O_2]^1} = [mol/L]^{-1}$

iii $K_c = \dfrac{[CaO]_s^1 [CO_2]^1}{[CaCO_3]_s} = \dfrac{[1][CO_2]^1}{[1]} = [CO_2]^1 = [mol/L]^1$

iv $2KCL_{3(s)} \xrightleftharpoons{\Delta} 2KCL_{(s)} + 3O_{2(g)}$

$K_c = \dfrac{[O_2]^3 \cdot 1}{1} = [O_2]^3$

$m_1A + m_2B \rightleftharpoons n_1C + n_2D$

$K_c = \dfrac{[C]^{n_1} . [D]^{n_2}}{[A]^{m_1} . [B]^{m_2}} = \dfrac{[mole/L]^{n_1} . [mole/L]^{n_2}}{[mole/L]^{m_1} . [mole/L]^{m_2}} = \dfrac{[mole/L]^{n_1+n_2}}{[mole/L]^{m_1+m_2}}$

units of $K_c = [mole/L]^{(n_1+n_2)-(m_1+m_2)}$

EQUILIBRIUM CONSTANT IN TERMS OF PRESSURE

- Liquids, solids are not taken into consideration

$$m_1A_{(g)} + m_2B_{(g)} \rightleftharpoons n_1C_{(g)} + n_2D_{(g)}$$

$$K_p = \frac{P_C^{n_1} \cdot P_D^{n_2}}{P_A^{m_1} \cdot P_B^{m_2}} = \frac{K_f}{K_b}$$

K_p → equilibrium constant (Pressure)

P_C, P_D, P_A, P_B are Partial pressures

units of K_p

$$K_p = (atm)^{(n_1+n_2)-(m_1+m_2)}$$

(n_1+n_2) → No. of moles of gaseous products

(m_1+m_2) → No. of moles of gaseous reactants

QUE:- Write K_p constants for the following reaction:-

i) $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$

ii) $CaCO_3(g) \rightleftharpoons CaO(s) + CO_2(g)$

iii) $2HI(g) \rightleftharpoons H_2(g) + I_2(g)$

Ans:-

i) $$K_p = \frac{(P_{PCl_3})^1 \cdot (P_{Cl_2})^1}{(P_{PCl_5})^1} = (atm)$$

iii $K_p = \dfrac{(P_{CO_2})^1}{(P_{CaCO_3})^1} = [M^0L^0T^0]$

iii $K_p = \dfrac{(P_{H_2})^1 (P_{I_2})^1}{(P_{HI})^2} = [M^0L^0T^0]$

iv $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$

$$K_p = \frac{(P_{NH_3})^2}{(P_{H_2})^3 \cdot (P_{N_2})^1} = (atm)^{-2}$$

Relationship Between K_p & K_c

$$\underset{(g)}{a_A} + \underset{(g)}{b_B} \rightleftharpoons \underset{(g)}{c_C} + \underset{(g)}{d_D}$$

$$K_p = \frac{P_C^c \cdot P_D^d}{P_A^a \cdot P_B^b} \quad \text{——— (1.)}$$

$$PV = nRT$$

$$P = \left(\frac{n}{V}\right) \cdot RT = C_M RT$$

eq (1) becomes

$$K_p = \frac{\{[C]RT\}^c \cdot \{[D] \cdot RT\}^d}{\{[A]RT\}^a \cdot \{[B] \cdot RT\}^b}$$

$$K_p = \frac{[C]^c \cdot [D]^d \cdot (RT)^{(c+d)-(a+b)}}{[A]^a \cdot [B]^b}$$

$$K_p = K_c (RT)^{\Delta n}$$

$$\Delta n = (c+d) - (a+b)$$

Write the relation between K_p and K_c :-

i. $N_{2(g)} + 3H_{2(g)} \rightleftharpoons 2NH_{3(g)}$

ii. $2SO_{3(g)} \rightleftharpoons 2SO_{2(g)} + O_{2(g)}$

iii. $CaCO_{3(s)} \rightleftharpoons CaO_{(s)} + CO_{2(g)}$

iv. $2HI_{(g)} \rightleftharpoons H_{2(g)} + I_{2(g)}$

i. $K_p = K_c (RT)^{\Delta n}$

$K_c (RT)^{(2)-(4)}$

$$\frac{[NH_3]^1}{[H_2]^3 [N_2]^1} \cdot (RT)^{-2} = K_c (RT)^{-2}$$

If $T = 273\ K$

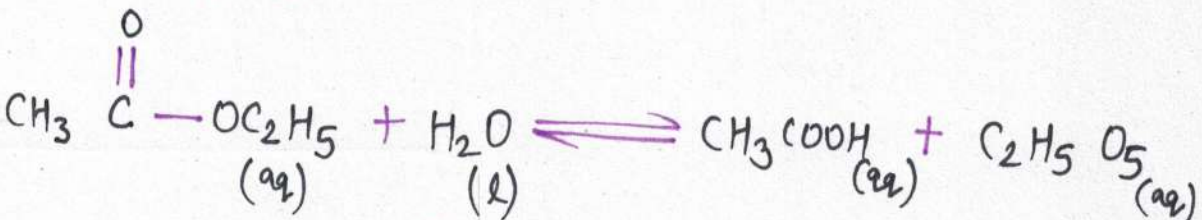
$K_p = \frac{K_c}{(22.4)^2}$; K_c ? K

ii. $K_p = \frac{[SO_2]^2 \cdot [O_2]^1}{[SO_3]^2} \cdot (RT)^1 = (K_c)(RT)$

iii. $K_p = [CO_2]^1 \cdot [RT]^1 = K_c (RT)$

IV $K_p = \frac{[H_2]^1 [I_2]^1}{[HI]^2} \cdot (RT)^0$

$= \frac{[H_2][I_2]^1}{[HI]^2} \cdot 1 = \underline{(K_c)}$

V 

$K_p = K_c (RT)^{2-2}$ — Not defined

$K_p = K_c$

But all are liquids

$\therefore K_p = K_c = 0$

K_p is not defined as none of the reactants and products are gaseous.

CHARACTERISTICS OF EQUILIBRIUM CONSTANT

- Equilibrium constant of a reaction depends only on "temperature". The temperature remains constant the equilibrium constant also remains same.
- Equilibrium constant doesn't depend on initial concentration of reactants and products.

Presence of an inert material doesn't change the equilibrium constant.
Presence of a material which reacts with either reactants or products changes the equation constant.

Presence of a catalyst doesn't change the equilibrium constant but in the presence of a catalyst the equilibrium is established quickly.

For the reactions, "$K >>>> 1$", equilibrium almost lies towards products and equilibrium conc^{n} of products are very high as compared to eq^{m} concentration of the reactants.

For the reactions, "$K <<<< 1$", the equilibrium lies towards the reactants and the equilibrium conc^{n} of reactants is very high as compared to products.

FACTORS AFFECTING EQUILIBRIUM CONSTANT

Mode of writing the reaction

$A_2 + B_2 \rightleftharpoons 2AB$

$$K_c = \frac{[AB]^2}{[A_2][B_2]}$$

$2AB \rightleftharpoons A_2 + B_2$

$$K_c' = \frac{[A_2][B_2]}{[AB]^2}$$

$$\therefore K_c' = \frac{1}{K_c}$$

If the reaction is reversed, the eq^m constant of the reversed reaction is equal to the reciprocal of the initial reaction.

ii Stichiometric co-efficients of the equilibrium reactions.

$$A_2 + B_2 \rightleftharpoons 2AB$$

$$K_c = \frac{[AB]^2}{[A_2].[B_2]}$$

Multiplying the eq^m with 2

$$2A_2 + 2B_2 \rightleftharpoons 4AB$$

$$K_c' = \frac{[AB]^4}{[A_2]^2.[B_2]^2}$$

$$K_c' = \left\{\frac{[AB]^2}{[A_2].[B_2]}\right\}^2$$

$$\therefore K_c' = K_c^2$$

If the reaction is multiplied with some no. then the equilibrium constant of the new reaction is equal to the eq^m constant of the initial reaction raised to a power of the number.

	Reaction (K_c)	K_c'
X	3	K_c^3
X	4	K_c^4
X	n	K_c^n

iii

Temperature

The only experimental factor that has an effect on the equilibrium constant is temperature. It can be explained by using "Vant Hoff's equation."

$$\log\left(\frac{K_2}{K_1}\right) = \frac{\Delta H}{2.303R} \cdot \left[\frac{1}{T^1} - \frac{1}{T^2}\right]$$

K_2 is eq^m const. at temp. T_2.

K_1 is eq^m const. at temp. T_1.

$\Delta H \rightarrow$ Enthalpy of the reaction.

- ΔH is -ve $\longrightarrow$ exothermic
- ΔH is +ve $\longrightarrow$ endothermic
- ΔH is 0 $\longrightarrow$ Thermo neutral

If $\Delta H = 0$;

$$\log\left(\frac{K_2}{K_1}\right) = 0$$

$$\therefore K_2 = K_1$$

Equilibrium constant of thermoneutral reaction doesn't depend on temperature.

If ΔH = -ve;

$T_2 > T_1$

$\frac{1}{T_1} - \frac{1}{T_2} = \frac{T_2 - T_1}{T_1 \cdot T_2}$ is a +ve quantity

$\therefore \Delta H$ is $-ve$;

$\log\left(\frac{K_2}{K_1}\right) < 0$

$\therefore$ $K_2 < K_1$

equilibrium constant of exothermic reaction decreases with the ↑ in temperature.

If $\Delta H = +ve$;

$T_2 > T_1$

$\frac{1}{T_1} - \frac{1}{T_2} = \frac{T_2 - T_1}{T_1 T_2}$ is $+ve$

$\therefore \Delta H$ is $+ve$

$\log\left(\frac{K_2}{K_1}\right) > 0$

$\therefore$ $K_2 > K_1$

equilibrium constant of endothermic reaction ↑ with ↑ in temperature

Equilibrium Constant of combined equilibrium reaction :-

$A \rightleftharpoons B$ $\quad K_1 = \frac{[B]}{[A]}$

$B \rightleftharpoons C$ $\quad K_2 = \frac{[C]}{[B]}$

$C \rightleftharpoons D$ $\quad K_3 = \frac{[D]}{[C]}$

$A \rightleftharpoons D \Rightarrow ?$

$\therefore$ on adding the eqⁿ;

$A + \cancel{B} + \cancel{C} \rightleftharpoons \cancel{B} + \cancel{C} + D \Rightarrow A \rightleftharpoons D$

$$\therefore K_4 = \frac{[D]}{[A]}$$

$$\frac{[D]}{[C]} \cdot \frac{[C]}{[B]} \cdot \frac{[B]}{[A]}$$

$2NO + O_2 \rightleftharpoons 2NO_2 \longrightarrow K_1$

$2NO + Br_2 \rightleftharpoons 2NO_2Br \longrightarrow K_2$

If the equation are added, then the eqⁿ constants are multiplied.

Calculate equilibrium constant of :-

(i) $2NO + O_2 \rightleftharpoons 2NO_2 \longrightarrow K_1$

$2NO + Br_2 \rightleftharpoons 2NO_2Br \longrightarrow K_2$ ②

$NO_2Br \rightleftharpoons \frac{1}{2}Br_2 + \frac{1}{2}O_2 + NO \longrightarrow K_3 = ?$

(i) $K_1 = \dfrac{[NO_2]^2}{[NO]^2[O_2]^1}$

$K_2 = \dfrac{[NO_2][Br_2]^2}{[NO_2]^2[Br_2]^1}$

$K_3 = K_1 \cdot K_2 = \dfrac{[NO_2]^2}{[NO]^2 \cdot [O_2]^1} \cdot \dfrac{[NO_2Br]^2}{[NO_2]^2[Br_2]}$

as we have multiplied by ½

$K_3 = \left(\dfrac{1}{K_1 \cdot K_2}\right)^{1/2} = \left\{\dfrac{[Br_2] \cdot [O_2] \cdot [NO]^2}{[NO_2][Br_2]^2}\right\}^{1/2}$

(as we have reversed the reaction)

After adding ① & ② ;

$$Br_2 + 2NO + O_2 \rightleftharpoons 2NO_2Br$$

$$2NO_2Br \rightleftharpoons Br_2 + O_2 + 2NO$$

$$NO_2Br \rightleftharpoons \tfrac{1}{2}Br_2 + \tfrac{1}{2}O_2 + NO$$

ii $\quad CoO_{(s)} + H_{2(g)} \rightleftharpoons Co_{(s)} + H_2O_{(g)}$ $\quad \hookrightarrow K_c = 67$

$$CoO_{(s)} + CO_{(g)} \rightleftharpoons Co_{(s)} + CO_2 \quad \hookrightarrow K_c = 490$$

Calculate:- K_c of →

$$CO_{2(g)} + H_{2(g)} \rightleftharpoons CO_{(g)} + H_2O_{(g)}$$

A ii $\quad CaO_{(s)} + H_{2(g)} \rightleftharpoons Co_{(s)} + H_2O_{(g)}$ $\quad \hookrightarrow 67$

$$CoO_{(s)} + CO_{(g)} \rightleftharpoons Co_{(s)} + CO_{2(g)} \quad \hookrightarrow 490$$

We need to cancel Co(s)

Therefore, reverse the reaction (2);

$$CoO_{(g)} + H_{2(g)} \rightleftharpoons Co_{(s)} + H_2O_{(g)} \quad \hookrightarrow 67$$

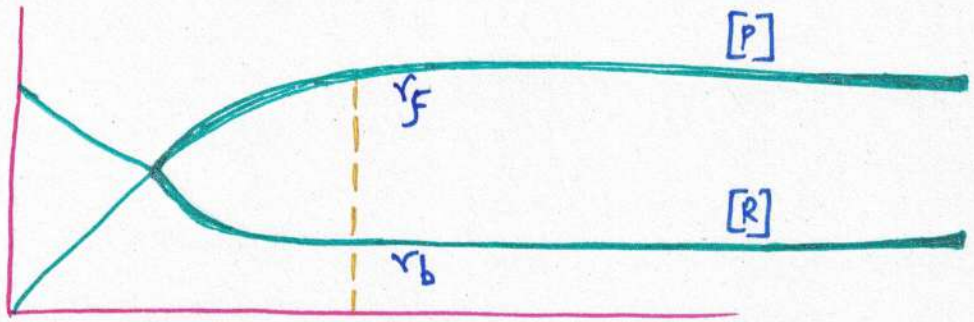
$$Co_{(s)} + CO_{2(g)} \rightleftharpoons CoO_{(s)} + CO_{(g)} \quad \hookrightarrow 1/490$$

$$CoO_{(g)} + H_{2(g)} + CO_{2(g)} \rightleftharpoons H_2O_{(g)} + CoO_{(s)} + CO_{(g)}$$

$$\therefore K_c = \frac{67}{490}$$

Write if the following graphs show equilibrium state and whether [R] = [P] ?

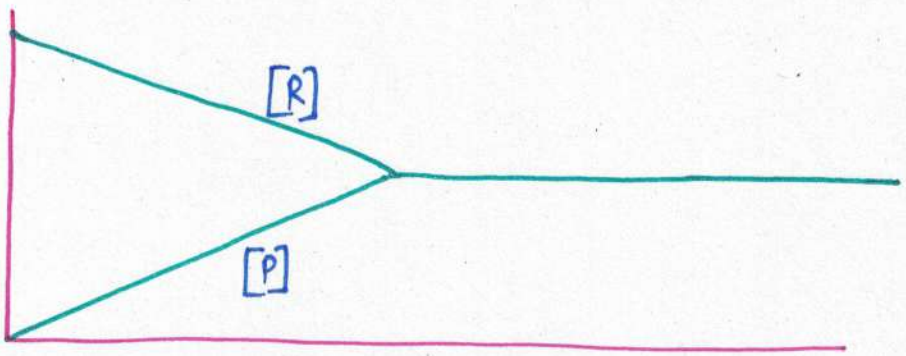
i



[R] = [P]

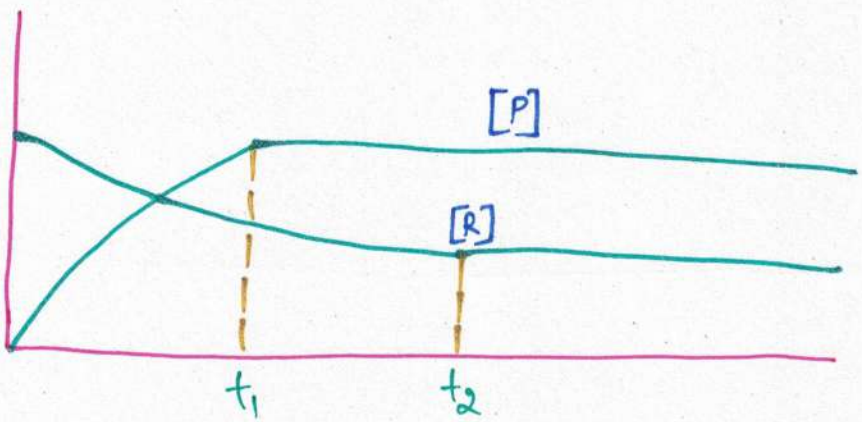
No equilibrium, no. overlapping of the rates

ii



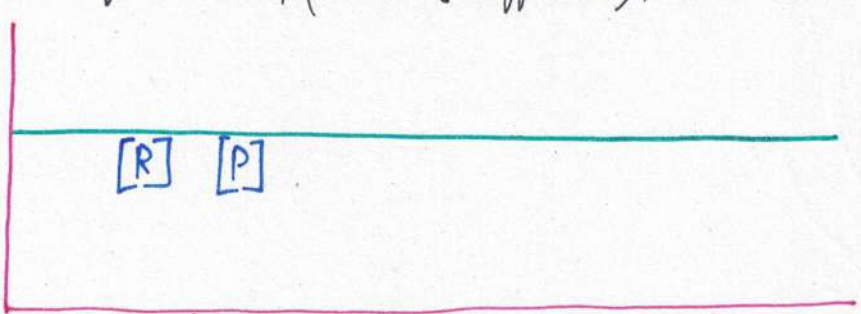
equilibrium is there, Conc^n is also same

iii



No equilibrium (time is different), Conc^n not same.

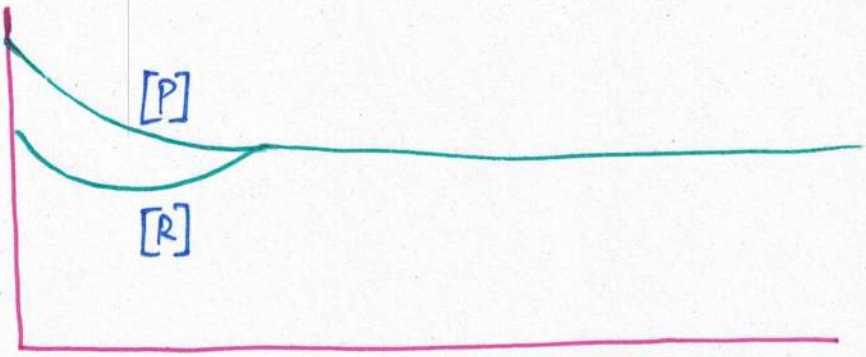
iv



Both the rates have been at eq^m from the beginning.

Conc^n is same.

V



No eq^{m} (Both rates can't be decreasing) Conc^{n} not same.

K_c and K_p are temperature dependent.



$A + B \rightleftharpoons C + D$ If initially A and B are in the ratio 2:1 at equilibrium conc^{n} of [B] and [D] are same? Calculate K_p & K_c?

Ans:- Q. $A + B \rightleftharpoons C + D$

$t=0$	$2a$	a		
$t=t$	$2a-x$	$a-x$	x	x

$$[B] = [D]$$

$$\frac{a-x}{V} = \frac{x}{V}$$

$$\therefore \boxed{x = \frac{a}{2}}$$

$$K_c = \frac{[C].[D]}{[B].[A]} = \left[\frac{a}{2V}\right] \Big/ \left[\frac{2a - a/2}{V}\right]$$

$$= \left[\frac{a}{2}\right] \Big/ \left[\frac{3a}{2}\right]$$

$$\boxed{K_c = \frac{1}{3} = K_p}$$

b. $A + B \rightleftharpoons C + D$ Ratio = 2:1 for A and B. At eq^m conc^n of B = Conc^n of D. find K_c = ?

Ans:- $A + 2B \rightleftharpoons C + D$

$t=0$	$(2a)$	(a)		
$t=t$	$(2a-x)$	$(a-2x)$	x	x

$$[B] = [D]$$

$$\frac{a-2x}{V} = \frac{x}{V}$$

$$x = \frac{a}{3}$$

$$K_c = \frac{[C][D]}{[A].[B].[B]} \cdot V$$

$$= \frac{[a/3].\ V}{\left[2a - \frac{a}{3}\right].\left[a - \frac{2a}{3}\right]}$$

$$K_c = \frac{3V}{5a}$$

c. $A + 2B \rightleftharpoons 2C + D$

Ratio = 2:1, $[B] = [D]$

Ans:- $A + 2B \rightleftharpoons 2C + D$

$t=0$	$2a$	a		
$t=t$	$2a-x$	$a-2x$	$2x$	x

$\Rightarrow [B] = [D]$

$a - 2x = x$

$$\boxed{x = \frac{a}{3}}$$

$$\Rightarrow \frac{\left[\frac{2a}{3}\right]^2 \cdot \left[\frac{a}{3}\right]}{\left[2a - \frac{a}{3}\right]\left[a - \frac{2a}{3}\right]^2}$$

$\Rightarrow 4/5$

DEGREE OF DISSOCIATION

This is the dissociation out of one mole.

$$\underset{a}{A_n} \rightleftharpoons nA$$

$a - x \qquad nx$

$a - a\alpha \qquad na\alpha$

$a \longrightarrow x$

$1 \longrightarrow x/a$

$\alpha = x/a \Rightarrow x = a\alpha$

$2A \rightleftharpoons B + C$

a

$a-2x$

$a \longrightarrow 2x$

$1 \longrightarrow 2x/a$

$\alpha = \frac{2x}{a} \Rightarrow x = \frac{a\alpha}{2}$

$a - a\alpha \longrightarrow n\left(\frac{a\alpha}{2}\right) + n\left(\frac{a\alpha}{2}\right)$

$(a(1-\alpha))$

?

$PCl_5 \rightleftharpoons PCl_3 + Cl_2$

If $\alpha = 0.3$ equilibrium pressure is 20 atm; what is the value of $K_p = ?$

A.

$PCl_5 \rightleftharpoons PCl_3 + Cl_2$

$t=0$	1	0	0
$t=1$	0.7	0.3	0.3

$$K_p = \frac{[PCl_3][Cl_2]}{[PCl_5]}$$

$$\frac{\left(\frac{0.3}{1.3} \times 20\right) \cdot \left(\frac{0.3}{1.3} \times 20\right)}{\left(\frac{0.7}{1.3} \times 20\right)}$$

$$K_p = \frac{180}{91} \text{ atm}$$

SHORTCUT

If α is provided we want to calculate K_p for the reaction, we will assume initial mole as unity.

If α is very small compared to unity, we can take;

$$(1-\alpha) \approx 1$$

$$(1+\alpha) \approx 1$$

If α is less than "0.07" then we consider it to be very small as compared to unity.



What will be K_p if degree of disassociation is small as compared to unity ? T_p at $eq^m = P$.

Ans:-

$$A_3B_2 \rightleftharpoons 3A + 2B$$

$$K_p = \frac{[A]^3 \cdot [B]^2}{[A_3B_2]^1}$$

$\alpha = 1$

$$A_3B_2 \rightleftharpoons 3A + 2B$$

$t=0$	1	0	0
$t=t$	$(1-\alpha)$	(3α)	(2α)

$(1-\alpha \approx 1)$

$$K_p = \frac{[A]^3[B]^2}{[A_3B_2]^1} \Rightarrow \frac{\left(\frac{3\alpha}{1+4\alpha} \cdot P\right)^3 \left(\frac{2\alpha}{1+4\alpha} \cdot P\right)^2}{\left(\frac{1-\alpha}{1+4\alpha} \cdot P\right)}$$

$$= \frac{27\alpha^3 p^3 \times 4\alpha^2 p^2}{P} \quad \{1-\alpha = 1\}$$

$$= 108\alpha^5 p_T^4$$



$$AB_2(g) \rightleftharpoons A(g) + 2B(g)$$

For the equilibrium K_p and total pressure P is given. If α is small as compared to unity, find α in terms of K_p and P?

Ans:-

	AB_2	$\rightleftharpoons$	A	+	$2B$
$t=0$	1		0		0
$t=t$	$(1-\alpha)$		α		2α

$$K_p = \frac{[A].[B]^2}{[AB]} = \frac{\left(\frac{\alpha}{1+\alpha}.P\right).\left(\frac{2\alpha}{1+\alpha}.P\right)^2}{\left(\frac{1-\alpha}{1+\alpha}.P\right)}$$

$$1-\alpha \approx 1 \qquad 1+\alpha \approx 1$$

$$\therefore K_p = 4\alpha^3 p^2$$

$$\boxed{\alpha = \left(\frac{K_p}{4P^2}\right)^{1/3}}$$

VAPOUR DENSITY CALCULATION FROM DEGREE OF DISSOCIATION

$$A_n \rightleftharpoons nA$$

	A_n	nA
$t=0$	a	0
$t=t$	$a(1-\alpha)$	$na\alpha$

Total moles $= a(1-\alpha) + na\alpha$

By mass conservation;

$$a.M_{A_n} = a\left[1+(n-1)\alpha\right]_{M_{mix}}$$

$$VD_{mix} = \frac{M_{mix}}{2}$$

$$M_{mix} = 2.VD_{mix}$$

SHORTCUT :- Take "a" as 1 because "a" will be cancelled out both sides.

?

$$PCl_{5(g)} \rightleftharpoons PCl_{3(g)} + Cl_{2(g)}$$

If $\alpha = 0.5$, then calculate the V.D of the mixture.

Ans:-

$$PCl_{5(g)} \rightleftharpoons PCl_{3(g)} + Cl_2$$

	PCl_5	PCl_3	Cl_2
$t=0$	1	0	0
$t=1$	0.5	0.5	0.5

Moles in total $= 1.5$

$$1.(208.5) = 1.(1.5).(VD_{mix}).2$$

$$VD_{mix} = \frac{208.5}{3} = 69.5\,g$$

Q $N_2O_4 \rightleftharpoons 2NO_2$, at 27°C, If the degree of dissociation is 0.2 calculate :-

1. K_p and K_c
2. V.D of mixture
3. What will be value of total pressure at equilibrium, if the volume is 2L ?

Ans:-

$$N_2O_4 \rightleftharpoons 2NO_2$$

$t=0 \quad 1 \rightleftharpoons 0$

$t=0 \quad 0.8 \rightleftharpoons 0.4$

$(1-\alpha) \quad (2\alpha)$

$t = 300K$

1. Total moles = 1.2

(a) (92) = a [1.2] . (VDmix) . 2

(92) = (VD) 2.4

$$(VD) = \frac{\cancel{920}^{\cancel{230}^{115}}}{\cancel{24}_{\cancel{6}_3}} = \frac{115}{3} = 38.3$$

2. $$K_p = \frac{(NO_2)^2}{(N_2O_4)^1} = \frac{(P_{NO_2})^2}{(P_{N_2O_4})^1}$$

$$= \frac{\left(\frac{2\alpha}{1+\alpha} \cdot P_T\right)^2}{\left(\frac{1-\alpha}{1+\alpha} P_T\right)}$$

$$K_p = \frac{4\alpha^2}{(1-\alpha)^2} \cdot P_T$$

$PV = nRT$

$P_T \cdot 2 = (1+\alpha) \cdot \frac{1}{12} \cdot 300^{25}$

$$P_T = \frac{25(1+\alpha)}{2}$$

$= \frac{25 \times 1.2^{0.6}}{2} = 15$ atm

$\Delta n = 1$

$\therefore K_p = \frac{4 \times 0.4 \times 15}{0.64 \cdot 4} = \frac{15}{4}$

$\frac{15}{4} = K_c \left(\frac{1}{12} \cdot \frac{300}{25}\right)$

$$K_c = \frac{3}{20}$$

$$K_p = \frac{15}{4}$$

3. $P_T = 15$ atm

SIGNIFICANCE OF REACTION QUOTIENT

$$aA + bB \longrightarrow cC + dD$$

$$Q_c = \frac{[C]^c \cdot [D]^d}{[A]^a \cdot [B]^b}$$

It is at general time "t"

i. If $Q_c = K_c \longrightarrow$ Equilibrium

ii. If $Q_c > K_c \longrightarrow$ Backward direction

iii If $Q_c < K_c$ ⟶ Forward direction



At any time of equilibrium $A + B \rightleftharpoons C + D$, each reactant and product were 2 moles. In which direction, will the eq^m move. if:-

i $K_c = 4$ ii 1 iii $K_c = 0.25$

Ans:- ii equilibrium

i $Q_c = \frac{[C]^2[D]^2}{[A]^2[B]^2} = 1$

$K_c = 4$ Forward

$Q_c < K_c$

iii $Q_c = 1$ $K_c = 0.5$ Backward



What will be equilibrium con^n in each case if vol. of container is 1L?

i $V = 1L$, $K_c = 4$, no. of moles = 2 each

	A	$+ B \rightleftharpoons$	C	$+ D$
$t=0$	2	2	2	2
$t=t$	$2-x$	$2-x$	$2+x$	$2+x$

$[A] = 2 - \frac{2}{3} = \frac{4}{3}M$ $\quad \frac{(2+x)^2}{(2-x)^2} = 4$

$[B] = \frac{4}{3}M$ $\quad \frac{2+x}{2-x} = 2$

$[C] = \frac{8}{3}M = [D]$ $\quad x = 2/3$

ii $A + B \rightleftharpoons C + D$

$[M] = 2$

iii $A + B \rightleftharpoons C + D$

	A	B	C	D
$t=0$	2	2	2	2
$t=t$	$2+x$	$2+x$	$2-x$	$2-x$

$$\frac{1}{4} = \left(\frac{2-x}{2+x}\right)^2$$

$$x = 2/3$$



$A + B \rightleftharpoons 2C$ at 300°C, moles of A, B, C are 2, 2, 3 moles respectively. If K_c at this temperature is 9; Volume = 1 L

i In which direction will the equilibrium move.

ii What will be equilibrium con^n of A, B, C ?

Ans:- i $Q_c = \frac{[C]^2}{[A][B]} = \frac{9}{4}$

$K_c = 9$

As $K_c > Q_c$

∴ The reaction will move in forward direction.

ii Equilibrium Con^n of $[A]$ =

$A + B \rightleftharpoons 2C$

	A	B	C
$t=0$	$2-x$	$2-x$	$3+2x$

$$\left(\frac{(3+2x)^2}{(2-x)^2}\right) = 9$$

$$x = 3/5$$

$A = 7/5$

$B = 7/5$

$C = 21/5$

(iii) If $\frac{4}{5}$ moles of C is added; what will be the new equilibrium Conc^n ? V = 1L.

Ans (iii)

$$\underset{7/5}{A} + \underset{7/5}{B} \rightleftharpoons \underset{\frac{21}{5}+\frac{4}{5}=5}{2C}$$

$$Q_c = \frac{[5]^2}{(7/5)^2} = \frac{25}{(7/5)\cdot(7/5)} > K_c$$

$$Q_c > K_c$$

∴ It will move in the backward direction

$$\underset{7/5+x}{A} + \underset{7/5+x}{B} \rightleftharpoons \underset{5-2x}{2C}$$

$$\Rightarrow \left(\frac{5-2x}{7/5+x}\right)^2 = 9$$

$$\Rightarrow (5-2x) = \left(\frac{7}{5}+x\right) 9$$

$$5(5-2x) = 7+5x \cdot 9$$

$$x = \frac{4}{25}$$

$$\therefore [A] = 7/5 + 4/25 = \frac{39}{25}\ M$$

$$[B] = 7/5 + 4/25 = 31/25\ M$$

$$[C] = \frac{125}{25} - 8/25 = \frac{117}{25} M$$

If Value of "K" is very large; $(K > 10^3)$

$$R \rightarrow P$$

$$K = \frac{[P]}{[R]}$$

$$[P] \gg [R]$$

$[P] + [R] \approx [P]$

$[P] - [R] \approx [P]$

We will solve question assuming forward reaction by limiting reagent concept.

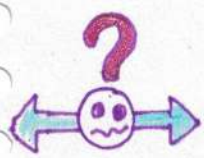
When value of "K" is very small; $(K < 10^{-3})$

$$R \longrightarrow P$$

$$K = \frac{[P]}{[R]}$$

$[R] \gg [P]$

$[R] \pm [P] \approx [R]$



$pH = -\log [H^+]$. What will be pH of 0.1M CH_3COOH ? Given equim constant for

$$CH_3COOH \rightleftharpoons CH_3COO^- + H^+$$

is $K_c = 10^{-5}$?

Ans:-

$$CH_3COOH \rightleftharpoons CH_3COO^- + H^+$$

$t=0$	0.1		
$t=t$	$0.1-x$	x	x

$$K = 10^{-5} = \frac{x^2}{0.1-x} = \frac{x^2}{0.1}$$

$$x = 10^{-3}$$

$$[x^+] = x = 10^{-3}$$

$$pH = -\log_{10} [10^{-3}]$$

$$\boxed{pH = 3}$$

$2O_3 \rightleftharpoons 3O_2$. If $K_p = 10^8$ At eqm total pressure is 10 atm. What will be partial pressure of each gas?

Ans:-

$$2O_3 \rightleftharpoons 3O_2$$

P 10

$$\frac{10^3}{P^2} = 10^8 \Rightarrow P = 10^{-5/2}$$

As K_p is very Large. Thus the rxn moves $P_{O_3} + P_{O_2} = 10$ atm towards forward directions.

ACTIVE MASS

- Concentration
- Partial pressure

Reaction are of two types :-

1. HOMOGENOUS REACTION

All reactants, products, catalysts ; all should be in same phase, All gases and aqueous solution, uniformly distributed system in single phase.

2. HETROGENOUS REACTION

Reactants, products, catalysts may be in two or more than two phases. These type of reaction are called hetrogeneous reaction.

All solids and liquids are in different phases.

eg:-

$$SO_{2(g)} + \tfrac{1}{2}O_{2(g)} \xrightleftharpoons{NO_{(g)}} SO_{3(g)}$$

Homogeneous

$$SO_{2(g)} + \tfrac{1}{2}O_{2(g)} \xrightleftharpoons{Pt_{(s)}} SO_{3(g)}$$

Hetrogeneous

$$CaCO_{3(s)} \rightleftharpoons CaO_{(s)} + CO_{2(g)}$$

Hetrogeneous

There are 3 phases because solids are taken in different phases along with CO_2 gas.

$$NH_4HS_{(s)} \rightleftharpoons NH_{3(g)} + H_2S_{(g)}$$

Hetro (2 phases)

$$CH_3COOH + CH_3CH_2OH \rightleftharpoons CH_3COOC_2H_{5(aq)} + H_2O_{(aq)}$$

same phase (aq.) ; Homogeneous

$$N_{2(g)} + 3H_{2(g)} \rightleftharpoons 2NH_{3(g)}$$

Homogeneous

$$N_{2(g)} + 3H_{2(g)} \rightleftharpoons 2NH_{3(g)}$$

? $N_{2(g)} + 3H_{2(g)} \rightleftharpoons 2NH_{3(g)}$

$t=0$	a	b	
$t=t$	$a-x$	$b-3x$	$2x$

$$K_c = \frac{\left[\frac{2x}{V}\right]^2}{\left[\frac{b-3x}{V}\right]^3\left[\frac{a-x}{V}\right]}$$

$$= \frac{\frac{4x^2}{V^2}}{[b-3x]^3 \cdot [a-x] \cdot \left[\frac{1}{V^2}\right]}$$

$$K_c = \frac{4x^2 V^2}{[b-3x]^3 [a-x]}$$

$$K_p = \frac{[P_C]^2}{[P_B]^3 \cdot [P_A]^1}$$

Total moles $= a+b-2x$

Mole fraction of $NH_3 = \frac{2x}{a+b-2x}$

Mole fraction of $H_2 = \frac{b-3x}{a+b-2x}$

Mole fraction of $N_2 = \frac{a-x}{a+b-2x}$

$$K_p = \frac{\left[\frac{2x}{a+b-2x}\right]^2}{\left[\frac{b-3x}{a+b-2x}\right]^3\left[\frac{a-x}{a+b-2x}\right]}$$

$$K_p = \frac{[4x^2][a+b-2x]^2}{(b-3x)^3 (a-x) P_T^2}$$

DEGREE OF DISSOCIATION

$$2NH_3 \rightleftharpoons N_2 + 3H_2$$

	$2NH_3$	N_2	$3H_2$
$t=0$	a		
$t=t$	$a-2x$	x	$3x$

$$a \longrightarrow 2x$$

$$1 \longrightarrow \frac{2x}{a} = \alpha$$

$$x = \frac{a\alpha}{2}$$

$$= a - 2\left(\frac{a\alpha}{2}\right) \qquad \frac{a\alpha}{2} \qquad \frac{3\alpha a}{2}$$

$$= a(1-\alpha) \qquad \frac{a\alpha}{2} \qquad \frac{3a\alpha}{2}$$

$$n_T = a(1-\alpha) + \frac{a\alpha}{2} + \frac{3a\alpha}{2}$$

$$= a(1+\alpha)$$

$$P_{NH_3} = \frac{a(1-\alpha)}{a(1+\alpha)} P_T$$

$$P_{N_2} = \frac{a\alpha/2}{a(1+\alpha)} P_T$$

$$P_{H_2} = \frac{3a\alpha/2}{a(1+\alpha)} P_T$$

$$\therefore K_p = \frac{(N_2)^1 (H_2)^3}{(NH_3)^2}$$

$$\frac{\frac{(\alpha/2)}{(1+\alpha)} \left(\frac{\alpha/2}{1+\alpha}\right)^3 P_T^4}{\left(\frac{1-\alpha}{1+\alpha}\right)^2 P_T^2}$$

$$K_p = \frac{27}{16} \cdot \frac{\alpha^4}{(1-\alpha^2)^2} P_T^2$$



$\alpha <<< 1$

$$\frac{1-\alpha}{1+\alpha} \simeq 1$$



"K" is small

$R \rightleftharpoons P$

$K < 10^{-3}$

$K_c = \frac{[P]}{[R]}$

$[R] >>> [P]$

$[R] + [P] \approx [R]$

$[R] - [P] \approx [R]$

"K" is Large

$R \rightleftharpoons P$

$K > 10^{3}$

$K_c = \frac{[P]}{[R]}$

$[P] >>> [R]$

$[R] + [P] \approx [P]$

$[R] - [P] \approx [P]$

Q

$2A + 3B \rightleftharpoons 3C + 2D$

Initially 3 moles of A, 8 moles of B, is given, If volume of container is 2L, value of K_c is 10^b, then calculate conc^n of [A], [B], [C], [D] = ?

	$2A$	+	$3B$	$\rightleftharpoons$	$3C$	+	$2D$
$t=0$	3		8				
$t=t$	$3-2x$		$8-3x$		$3x$		$2x$

Value of K_c is very large $> 10^3$

$[P] \gg [R]$

$[A] = \left[\frac{3-2x}{2}\right]^2$

$[B] = \left[\frac{8-3x}{2}\right]^3$

$[C] = \left[\frac{3x}{2}\right]^3$

$[D] = \left[\frac{2x}{2}\right]^2$

Ans:-

	$2A$	+	$3B$	$\rightleftharpoons$	$3C$	+	$2D$
$t=0$	3		8				
	L.R = A						
$t=t$	x		$8-9/2$		$9/2$	=	3
			$= 7/2$		$9/2$		3

$$10^b = \frac{(9/2)^3 \cdot (3)^2}{(7/2)^3 \cdot (x)^2}$$

$$x = \sqrt{\frac{(9/2)^3 \cdot (3)^2}{(7/2)^3} \times 10^{-b}}$$

$$x = \sqrt{\frac{27 \times 9}{343} \cdot 10^{-b}}$$

$$x = \frac{27}{7} \times \frac{3}{\sqrt{7}} \times 10^{-3}$$

For Calculation of Conc^n of L.R ; value of K_c will be provided, when K_c is Large.

$[C] = 9/2 \times 1/2\ M$

$[D] = 3/2\ M$

$[B] = 7/2 \times 1/2 = 7/4\ M$

$[A] = \frac{27 \times 3}{7\sqrt{7}} \times 10^{-3}\ M$ → small

HETROGENOUS EQUILIBRIUM

$$CaCO_{3(s)} \rightleftharpoons CaO_{(s)} + CO_{2(g)}$$

- Density of solids is constant at constant temperature.
- On increasing temperature, the density of solids as well as liquids will decrease due to volume expansion.

$$\text{Conc}^n = \frac{\text{Mole}}{\text{Volume}}$$

$$\text{Mole} = \frac{\text{Given mass}}{\text{Mole. mass}}$$

$$\therefore \text{Conc}^n = \frac{\text{Mass}}{\text{M.V}}$$

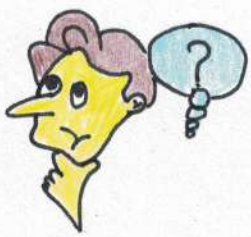
$$\text{Conc}^n = \frac{d}{M}$$

Therefore, concentration of the solids remains constant as density as well as molecular weight remains constant.

$$CaCO_{3(s)} \rightleftharpoons CaO_{(s)} + CO_{2(g)}$$

$$K = \frac{[CaO].[CO_2]}{[CaCO_3]}$$

$$\boxed{K_c = [CO_2]}$$



200g $CaCO_3$ is taken in a 1L container At 300K, 50% $CaCO_3$ gets dissociated a/c to equilibrium.

$$CaCO_{3(s)} \rightleftharpoons CaO_{(s)} + CO_{2(g)}$$

i) What will be the value K_p and K_c ?

ii) After getting equilibrium, if we'll get add 50g $CaCO_3$, what will be conc^n of CO_2 ?

$$CaCO_{3(s)} \rightleftharpoons CaO_{(s)} + CO_{2(g)}$$

t=0	2		
t=t	2-x	x	x
	2-1	1	1

i)

$$K_c = [CO_2]$$

$$K_c = 1\ mol/l$$

(ii) No change

$P = C \cdot R \cdot T$

$P = 1 \times \frac{1}{12} \times 300$ $\{[CO_2] = 1\}$

$K_p = 25$ atm

(iii) If Conc^n of CO, is 1 mol/l in the above equilibrium, in which direction will eq^m go when we add 50g CaO solid?

No change in the pathway.

(iv) What will happen when this eq^m mixture is put into 1.5 l container?

No change in the K_c as $K_c \propto T$.

$$CaCO_{3(s)} \rightleftharpoons CaO_{(s)} + CO_{2(g)}$$

$K_c = 1$

$Q_c = \frac{1}{3/2} = 2/3$

$Q_c < K_c$

∴ It goes in forward direction.

(v) How much amount is gone forward?

$$\underset{1-x}{CaCO_{3(s)}} \rightleftharpoons \underset{1+x}{CaO_{(s)}} + \underset{1+x}{CO_{2(g)}}$$

$\left[\frac{1+x}{1.5}\right] = 1$ {As K_c only depends on $[CO_2]$}

$$\underset{1/2}{CaCO_{3(s)}} \rightleftharpoons \underset{3/2}{CaO_{(s)}} + \underset{3/2}{CO_{2(g)}}$$

VI When eqm mixture is put into 0.5 l mixture?

$$CaCO_{3(s)} \rightleftharpoons CaO_{(s)} + CO_{2(g)}$$

It goes in the backward direction

$$\underset{1+x}{CaCO_{3(s)}} \rightleftharpoons \underset{1-x}{CaO_{(s)}} + \underset{1-x}{CO_{2(g)}}$$

$$\left[\frac{1-x}{1/2}\right] = 1$$

$$x = 1/2$$

$$\therefore \underset{3/2}{CaCO_{3(s)}} \rightleftharpoons \underset{1/2}{CaO_{(s)}} + \underset{1/2}{CO_{2(g)}}$$

VII When eqm mixture is put into 3L container? Pressure = ?

$$CaCO_{3(s)} \rightleftharpoons CaO + CO_{2(g)}$$

The eqm goes into the forward direction.

$$\underset{1-x}{CaCO_{3(s)}} \rightleftharpoons \underset{1+x}{CaO_{(s)}} + \underset{1+x}{CO_{2(g)}}$$

$$\left[\frac{1+x}{3}\right] = 1$$

$$\boxed{x = 2}$$

This reaction is not possible

$\therefore$ Max x possible $x = 1$

$CaCO_{3(s)} \rightleftharpoons CaO_{(s)} + CO_{2(g)}$

0 2 2

eqm is finished

For CO_2 gas

$PV = nRT$

$P = \frac{2}{3} \times \frac{1}{12} \times \frac{50}{300}$

$P = \frac{50}{3}$ atm

SHORTCUT

- $NH_4HS_{(s)} \rightleftharpoons NH_{3(g)} + H_2S_{(g)}$

$P_T = P$

$P/2$ $P/2$

$K_p = \frac{P}{2} \times \frac{P}{2} = \frac{P^2}{4}$

- $A_2B_{(s)} \rightleftharpoons 2A_{(g)} + B_{(g)}$

$2P/3$ $P/3$

$K_p = \left(\frac{2P}{3}\right)^2 \cdot \left(\frac{P}{3}\right)$

$K_p = \frac{4P^3}{27}$

Question:- $A_2B_3 \rightleftharpoons 2A_{(g)} + 3B_{(g)}$

$P_T = P$ $\quad K_p = ?$

Answer:- $\left(\frac{2P}{5}\right)^2 \cdot \left(\frac{3P}{5}\right)^3$

$= \frac{27 \times 4}{25 \times 125} \times P^5$

$$\boxed{K_p = \frac{108}{3125} \times P^5}$$

Revision

Chemical equilibrium

$N_2O_4 \rightleftharpoons 2NO_2$ $\quad \alpha = 0.5$ find $V.D_{min.} = ?$

Ans:- $N_2O_4 \rightleftharpoons 2NO_2$

$t=0$	1	
$t=t$	$1-\alpha$	2α

$1 \times 92 = (1+\alpha) \cdot M_{min}$

$\frac{92}{1.5} = M_{min}$

$\therefore \boxed{V.D_{mix} = \frac{92}{3}}$

Qu:- $N_2O_4 \rightleftharpoons 2NO_2$ Initial pressure in container = 600 atm, find pressure = 800 atm, find K_p = ?

Ans:- As volume and temp. is fixed, therefore $P \propto n$, we can assume the initial mole as 600.

$$N_2O_4 \rightleftharpoons 2NO_2$$

t=0	600	
t=t	600-p	2p → 400 atm

→ 400 atm

$$600 - p + 2p = 800$$

$$P = 200 \text{ atm}$$

$$\therefore K_p = \frac{P_{NO_2}}{P_{N_2O_4}} = \frac{(400)^2}{400} = 400 \text{ atm.}$$

Le Chatelier's Principle



If chemical reaction is at eqm; on applying strain (change in pressure/volume/temp. etc); the equilibrium will oppose that strain and goes in that direction in which strain is nullified.

Effect of concentration – An increasing conc^n eqm will shift towards opposite direction.

$$A \rightleftharpoons B$$

$$K_c = \frac{[B]}{[A]}$$

i On increasing [A];

i.e $[A]' > [A]$

$Q_c < K_c$

forward direction

ii on increasing [B];

i.e $[B]' > [B]$

$Q_c > K_c$

Backward direction

Effect of Volume

- on increasing volume, eqm will shift towards number of gaseous moles are more.

i $A \rightleftharpoons B + C$

a b c

$$K_c = \frac{bc}{av}$$

$$Q_c = \frac{bc}{av'}$$

$v' > v$

$Q_c < K_2$

Forward direction

ii $A + B \rightleftharpoons C$

a b c

$$K_c = \frac{cv}{ab}$$

$$Q_c = \frac{cv'}{ab}$$

$v' > v$

$Q_c > K$

Backward direction

$\Delta n > 0$ Forward

$\Delta n < 0$ Backward

$\Delta n = 0$ No effect

Effect of Pressure

- on increasing pressure on eqm, it will shift towards that direction where no. of gaseous moles are less.

Equilibrium is established at constant temperature. So, we can apply Boyl's Law.

$$P \propto \frac{1}{V}$$

$\Delta n > 0$ Backward
$\Delta n < 0$ Forward
$\Delta n = 0$ No effect

eg:-

$$A_{(g)} + 2B_{(g)} \rightleftharpoons 3C_{(g)} + \Delta_{(g)}$$

$\Delta n = 4 - 3 = 1$
$\Delta n > 0 \longrightarrow$ Backward

(gas which is non reactive with eqm)

Effect of Addition of Inert gas

- Inert gas is a gas which is non reactive with eqm.

a. At Constant volume

There will be no effect at equilibrium.

Let us suppose in a close container (A, B, C) are in equilibrium.

$$A \rightleftharpoons B + C$$

t=t a b c

$$K_p = \frac{P_B \cdot P_C}{P_A} \; ; \quad P_T = (a+b+c) \cdot \frac{RT}{V}$$

$$P_A V = aRT$$

$$P_B V = bRT$$

$$P_C V = cRT$$

Now add an inert gas "D" (moles = "d"), then total pressure will increase but partial pressure of individual gases remains same.

$$P_T = P_A + P_B + P_C + P_D$$

$$P_T \cdot V = (a+b+c+d)\,RT$$

$$P_T = (a+b+c+d) \cdot \frac{RT}{V}$$

$$P_A = \frac{a}{a+b+c+d} \cdot P_T$$

$$P_A = \frac{a}{(a+b+c+d)} \cdot (a+b+c+d) \cdot \frac{P_T}{V}$$

$$\boxed{P_A = \frac{aRT}{V}}$$ Similarly. For all.

∴ There is no effect of adding inert gas at const. volume.

b. At Constant Pressure

Effect will be same as increase in volume.

i.e:- $\Delta n > 0$ forward

$\Delta n < 0$ backward

$\Delta n = 0$ no effect

Effect of Catalyst

- catalyst are substances which can accelerate or retard the chemical reaction OR Catalyst can make rate of reactions faster or slower.

There will be no effect at equilibrium conc^{n} on addition of catalysts. eq^{m} may be reached earlier with the help of catalyst.

Effect of Temperature

Types of reactions :-

(i) Exothermic Reaction

The Reaction in which heat is ejected are called exothermic reaction.

$$A \rightleftharpoons B + q\,Cal$$

$$\Delta H < 0$$

(ii) Endothermic Reaction

The Reaction in which heat is absorbed.

$$A + q\,Cal \rightleftharpoons B$$

$$\Delta H > 0$$

Gibb's Free energy

$$\Delta G = \Delta H - T.\Delta S$$

ΔH = Enthalpy

T = Temp. of reaction

ΔS = change in entropy

Standard Conditions

25°C temperature, 1 atm pressure, concentration of species = 1M

$$\Delta G^{\circ} = \Delta H^{\circ} - T.\Delta S^{\circ}$$

At Standard Cond^{n} Gibb's free energy

$$\Delta G = \Delta G^\circ + RT \ln Q$$

ΔG = Gibb's free energy at a temperature
ΔG° = Gibb's free energy at a standard
Q = Reaction Quotient

$\Delta G < 0$ system (process) is feasible
$\Delta G > 0$ Process is non feasible
$\Delta G = 0$ process is at equilibrium

At equilibrium,

$\Delta G = 0$;

$\Delta G^\circ + RT \ln Q = 0$

$-RT \ln Q = \Delta G^\circ$ —— (1.)

$\Delta G^\circ = \Delta H - T.\Delta S^\circ$

From (1.)

$-RT \ln K_c = \Delta H^\circ - T.\Delta S^\circ$

{ At equilibrium $K_c = K_p = Q_c$ }

Conc^n of the products is more than the conc^n of the reactants; thus, the equilibrium will shift towards endothermic direction.

eg:- (i) $A \rightleftharpoons B$ $\Delta H > 0$

It's endothermic on increasing temp, eq^m reaction goes in the forward direction.

(ii) $A \rightleftharpoons B$ $\Delta H < 0$

It's exothermic on increasing temp, eq^m shifts in the backward direction.

Ex:- $N_{2(g)} + 3H_{2(g)} \rightleftharpoons 2NH_{3(g)}$ (ΔH -ve)

Δn is negative

∴ The reaction goes in backward direction originally but we want more production of NH_3. Therefore, the volume must be decreased pressure must be increased, temperature must be Lowered, Conc^n of N_2 and H_2 must be high so that the reaction moves in the forward direction.

PHYSICAL EQ^M

In this Section, we'll discuss eq^m in which physical state will be changed.

Melting of Solids

Solid $\rightleftharpoons$ Liquid

There are two types of solids

(i) $d_{solids} < d_{Liquids}$ eg:- Water and ice

(ii) $d_{solids} > d_{Liquids}$ eg:- Any Solid

a. WATER

Ice $\rightleftharpoons$ Water

$\Delta H > 0$

Volume of Ice will be greater than V_{H_2O}, for same mass because $d_{ice} < d_{H_2O}$
Hence, on high pressure eq^m will shift towards right. This is endothermic

reaction, so, on high temp, eqm will shift towards right.

Melting of ice will be favourable on high temp and high pressure.

$$\text{Ice} \rightleftharpoons \text{Water} \qquad \Delta H > 0$$

on increasing temp, we get more H_2O.

$$P \propto \frac{1}{V}$$

Volume of Ice is more, therefore pressure must be high (vol. of H_2O is less)

For Vol, temp ↑

b. SOLID

$$A_{(s)} \rightleftharpoons A_{(l)} \qquad (\Delta H > 0)$$

For solid A, density in liquid state will be less, so vol. of A in its liquid state will be more. so eqm will shift towards right on Low pressure and high temp.

c. LIQUID ⇌ VAPOUR eqm

(i) Vapour Pressure

It is the pressure exerted by the vapour on its own liquid when liquid is in eqm with vapour.

Vapour Pressure only depends on temperature

$$\log\left(\frac{P_2}{P_1}\right) = \frac{\Delta H}{2.303\,R}\left(\frac{1}{T_1} - \frac{1}{T_2}\right)$$

Where P_1 is V.P at temp. T_1 and P_2 is V.P at temp. T_2.

Vapour Pressure doesn't depend on amount of Liquid.

Question:- To an evacuated vessel with a movable piston under external pressure of 1 atm 0.1 mole of He and 1 mole of unknown compound (V.P = 0.68 atm of 0°C) are introduced, considering the ideal gas behaviour find total volume of gaseous mixture of 0°C.

Answer:- For unknown compound

$$0.68 \times V = 1 \times \frac{1}{12} \times 273$$

$$V = \frac{273 \times 100}{68 \times 12}$$

For He gas,

$$P = 0.32,\ V = ?,\ n = 0.1,\ R = 1/12,\ T = 273$$

$P_T = 1$ atm

$\therefore P_{He} = 0.32$ atm

$$0.32 \times V = 0.1 \times \frac{1}{12} \times 273$$

$$V = \frac{22.4 \times 0.1}{0.32} = 7\text{l}$$

Question:- A gaseous mixture of He and $H_2O_{(g)}$ in a V L container has a total pressure of 2.1 atm (V.P of H_2O is 0.1 atm). Find the total pressure of gaseous mixture if it is completelly transfered to :-

(i) V/3 L

(ii) 3V L

Answer:- Pressure of water will remain constant when we change the vol. of the container only the pressure exerted by "He" gas changes.

(i) $P_1V_1 = P_2V_2$

$2 \times V = P_2 \times V/3$

$$\boxed{P_2 = 6 \text{ atm}}$$

$\therefore T_p = P_2 + 0.1 = 6.1$ atm

(ii) $P_1V_1 = P_2V_2$

$2 \times V = (3V) \cdot P_2$

$$\boxed{P_2 = \frac{2}{3}}$$

$\therefore T_P = \frac{2}{3} + 0.1 \text{ atm}$

VP only depend on temperature

Relative Humidity

$$RH = \frac{\text{V.P of water}}{\text{Saturated V.P}}$$

Pressure of Dry air } $P_{\text{moist air}}$ – Aqueous tension

$P_{\text{moist air}}$ – VP_{water}

Ans:- $P_{sat} = \frac{22.8}{760}$

$P_{H_2O} = 10^{-2}$ atm

$R_H = \frac{10^{-2} \times 760}{22.8}$

$R_H = 1/3$

To $CuSO_4 \cdot 5H_2O$ to lose water, the eq^m must go in the forward direction.

$$\boxed{R_H < 1/3}$$

Solubility of a gas in Liquid

$$A_{(g)} + B_{(l)} \rightleftharpoons A_{(aq)}$$

$\Delta H < 0$, If temp ↑, it will go in backward direction; therefore, solubility ↓

Henry's Law :- Solubility of a gas in a liquid depends on the Partial Pressure of the gas over the Liquid.

$$x_{gas} = \frac{n_g}{n_g + n_l} \quad \{\text{As } n_g \text{ is very small}\}$$

$$P_{gas} \propto x_{gas}$$

$$P_{gas} = K_H \cdot x_{gas}$$

$$\therefore x_{gas} \approx \frac{n_g}{n_l}$$

$$\therefore P_{gas} = K_H \cdot \frac{n_g}{n_l}$$

Question

The Henry constant of solubility of N_2 in H_2O at 298K is 10^5 atm. The mole fraction of N_2 in air is 0.8. Find out the amount of N_2 dissolved in 10 mole of H_2O at 5 atm.

Answer

$K_H = 10^5$, $T = 298\ K$

$n_{N_2} = 0.8$, $n_{H_2O} = !$

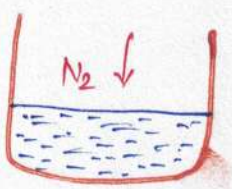
It can't be used as it is in air, Not dissolved

Total mole of $H_2O = 10$ mol

$P = 5$ atm $\quad n_{N_2} = 0.8$

$\therefore P_{N_2} = 0.8 \times 5 = 4$ atm

$$P_{N_2} = K_H = \frac{n_{N_2}}{10}$$



$$4 = 10^5 \cdot \frac{n_{N_2}}{10}$$

$$\therefore n_{N_2} = \frac{4}{10^4}$$